Practical 6: Numeric Data

Easing into EDA with Pandas

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This session is a tour-de-`pandas`; since this is Python's equivalent of the `tidyverse` meets `data.tables` it is fundamental to the data science ecosystem and is probably one of the most-widely used libraries in the language as a whole. I get more than 286,000 questions tagged with pandas on StackOverflow.

This week we are also going to start looking more closely at the **InsideAirbnb** data which forms the core of the work that we do over the rest of the term. The focus of *this* notebook is simple numeric data: no mapping or text data… yet… and direct manipulation of data types, derivation of summary statistics, and simple plotting.

We hope that you will be able to draw on the past few practical sessions to develop a more intuitive understanding of how to interact with pandas since it supports both a 'dictionary-of-lists' style of interaction *and* a methods-based style of interaction with the 'Data Frame'.

> ⚠ Important
>
> Conceptually, this practical links together *all* of the preceding ones; you will find data structures, classes and methods, reading CSV files from a remote location, `numpy`, and more than you ever wanted to know about data types in Python. Making these connections will make the remainder of term much, much easier, so it might be worth **revising this practical** over Reading Week so make sure it all makes sense! That may include rewatching the lectures on Data and Pandas

1 The Importance of EDA

After a few weeks getting to grips with Python, we're now going to start working with some real data. One of the first things that we do when working with any new data set is to familiarise ourselves with it. There are a *huge* number of ways to do this, but there are no shortcuts to:

1. Reading about the data (how it was collected, what the sample size was, etc.)
2. Reviewing any accompanying metadata (data about the data, column specs, etc.)
3. Looking at the data itself at the row- and column-levels
4. Producing descriptive statistics
5. Visualising the data using plots

You should use *all* of these together to really understand where the data came from, how it was handled, and whether there are gaps or other problems. If you're wondering which comes first, the concept of *start with a chart* is always good… though we've obviously not *quite* gotten there yet! This week we want you to get a handle on pandas itself, so although we will do some plotting of charts, we'll focus on 3-4 with a tiny bit of 5. There will be much more on plotting charts next week, and you should be looking into 1 and 2 yourself based on what's been written both on the Inside Airbnb web site and in the suggested readings.

So although they don't need to be done now, you probably want to add both those links to your reading list!

2 Pandas, Conceptually

> ℹ Connections
>
> This is why we spent time talking about Packages, Methods Classes… because now we're going to be making *intensive* use of them.

Pandas can do a *lot*, and you might be feeling a little intimidated by this, but here's the thing: we were already writing something like pandas from scratch! That's because pandas takes a **column-view of data** in the same way that our **Dictionary-of-Lists** did, it's just that it's got a lot more features than our 'simple' tool did. That's why the documentation is so much more forbidding and why pandas is so much more powerful.

But at its heart, a pandas `Data Frame` (`df` for short) is a collection of `Data Series` objects (i.e. columns) with an index. Each Series is like one of our column-lists from the last notebook. And the `df` is like the dictionary that held the data together. So you've seen this before and you

already *know* what's going on… or at least you now have an *analogy* that you can use to make sense of pandas:

```
myDataFrame = {
    '<column_name_1>': <Series_1>,
    '<column_name_2>': <Series_2>,
    '<column_name_3>': <Series_3>
}
```

And pandas gives us two ways to access that data:

1. Using a method syntax: `myDataFrame.column_name_1`
2. Using a dictionary syntax: `myDataFrame['column_name_1']`[1]

Depending on which syntax you prefer, you can use these interchangeably. The only times you *have* to choose one over the other are:

- Assignment (e.g. `myDataFrame['column_name_1'] = ...`);
- Columns with spaces in their names (e.g. `myDataFrame['Column Name 1')`).

There are numerous useful introductions; one of our favourites is from Greg Reda, and there are some good videos on our YouTube channel. And of course, there's TONS of stuff on StackOverflow. If you want an actual physical book, you might try McKinney (2017).

3 Reading Data

It's always sensible to import the packages you need at the top of the notebook:

1. Because it lets everyone know what they need to have installed to run your code.
2. It's easy to run this and then skip further down the notebook if you have already done *some* of the work and saved an intermediate output.

So in this practical we need:

```
from pathlib import Path
import numpy as np
import pandas as pd
```

One thing you will really want to bookmark is the official documentation since you will undoubtedly need to refer to it fairly regularly. *Note*: this link is to the most recent release. Over time there will be updates published and you *may* find that you no longer have the most up-to-date version. If you find that you are now using an older version of pandas and the methods have changed then you'll need to track down the *specific* version of the documentation that you need from the home page.

You can always check what version you have installed like this:

```
print(pd.__version__)
```

```
3.0.6
```

[1] This is also how `[polars](https://pola.rs/)` does things

> **Tip**
>
> The `<package_name>.__version__` approach isn't guaranteed to work with *every* package, but it will work with most of them. Remember that variables and methods starting and ending with '__' are **private** and any interaction with them should be approached very, very carefully.

3.1 Reading Remote Data

> **Difficulty: Low (this time around).**

You will need to do several things here to read the remote, compressed CSV file specified by `url` into a data frame called `df`. Setting `low_memory=False` ensures that pandas will try to load the entire data set *before* guessing the data type for each column! Obviously, with very large files this is probably a bad idea and it's possible to force a particular column type while readng in the data as well.

> For larger data sets there is Polars, DuckDB, and platforms like Dask (see, eg, this) and beyond.

3.1.1 Parameterising Files

Anyway, here's where to find the data:

```
ymd  = '20260619'
city = 'London'
host = 'http://orca.casa.ucl.ac.uk'
url  = f'{host}/~jreades/data/{ymd}-{city}-listings.csv.gz'
```

3.1.2 Reading CSV with Pandas

Now, what goes into the `read_csv` function below?

Question

```
# your code here
df = pd.read_csv(??, compression='gzip', low_memory=False)
print(f"Data frame is {df.shape[0]:,} x {df.shape[1]}")
```

You should get a data frame containing 79 columns and 96,651 rows of data.

4 Inspecting the Data Frame

Difficulty: Low.

Let's get a general sense of the data by printing out information *about* the data frame. There are several ways to do this (and we'll see another futher on):

- `df.describe(percentiles=None, include=None, exclude=None, datetime_is_numeric=Fals` – descriptive stats for all **numeric** columns
- `df.info(verbose=None, buf=None, max_cols=None, memory_usage=None, show_counts=None)` – summarises all columns, but without distribution information
- `df.memory_usage(index=True, deep=True)` – memory usage details about each column (can be quite slow as it's doing a *lot* of digging)

Question

What is another term for the 0.5 percentile?

4.0.1 Describing

Describing a data frame provides general information about *numeric* columns, such as the median, IQR, or number of discrete values.

So to show the 5th and 95th percentiles you need to pass an argument to `describe` to override the default report from pandas. Notice here how we can **subselect columns** using a list: `df[[<names of selected columns>] ]`!

Question

```
df[
    ['latitude', 'longitude', 'accommodates', 'bathrooms', 'bedrooms', 'beds']
].describe(percentiles=[??])
```

4.0.2 Info

The `info` method provides a more system-oriented view of the data frame, helping you to understand what each column is composed of, how many NAs there might be, and some high-level (but often incomplete) data on performance.

```
df.info(verbose=True)
```

```
<class 'pandas.DataFrame'>
RangeIndex: 96651 entries, 0 to 96650
Data columns (total 79 columns):
 #   Column                                        Non-Null Count  Dtype
---  ------                                        --------------  -----
 0   id                                            96651 non-null  int64
```

```
1   listing_url                                    96651 non-null  str
2   scrape_id                                      96651 non-null  int64
3   last_scraped                                   96651 non-null  str
4   source                                         96651 non-null  str
5   name                                           96651 non-null  str
6   description                                    93806 non-null  str
7   neighborhood_overview                          41983 non-null  str
8   picture_url                                    96642 non-null  str
9   host_id                                        96651 non-null  int64
10  host_url                                       96651 non-null  str
11  host_name                                      96611 non-null  str
12  host_since                                     96613 non-null  str
13  host_location                                  73323 non-null  str
14  host_about                                     49314 non-null  str
15  host_response_time                             63627 non-null  str
16  host_response_rate                             63627 non-null  str
17  host_acceptance_rate                           69036 non-null  str
18  host_is_superhost                              94828 non-null  str
19  host_thumbnail_url                             96613 non-null  str
20  host_picture_url                               96613 non-null  str
21  host_neighbourhood                             46291 non-null  str
22  host_listings_count                            96613 non-null  float64
23  host_total_listings_count                      96613 non-null  float64
24  host_verifications                             96613 non-null  str
25  host_has_profile_pic                           96613 non-null  str
26  host_identity_verified                         96613 non-null  str
27  neighbourhood                                  41984 non-null  str
28  neighbourhood_cleansed                         96651 non-null  str
29  neighbourhood_group_cleansed                   0 non-null      float64
30  latitude                                       96651 non-null  float64
31  longitude                                      96651 non-null  float64
32  property_type                                  96651 non-null  str
33  room_type                                      96651 non-null  str
34  accommodates                                   96651 non-null  int64
35  bathrooms                                      62730 non-null  float64
36  bathrooms_text                                 96502 non-null  str
37  bedrooms                                       84071 non-null  float64
38  beds                                           62676 non-null  float64
39  amenities                                      96651 non-null  str
40  price                                          62684 non-null  str
41  minimum_nights                                 96651 non-null  int64
42  maximum_nights                                 96651 non-null  int64
43  minimum_minimum_nights                         96651 non-null  int64
44  maximum_minimum_nights                         96651 non-null  int64
45  minimum_maximum_nights                         96651 non-null  int64
46  maximum_maximum_nights                         96651 non-null  int64
47  minimum_nights_avg_ntm                         96651 non-null  float64
48  maximum_nights_avg_ntm                         96651 non-null  float64
49  calendar_updated                               0 non-null      float64
50  has_availability                               92188 non-null  str
51  availability_30                                96651 non-null  int64
52  availability_60                                96651 non-null  int64
```

```
 53  availability_90                                 96651 non-null  int64
 54  availability_365                                96651 non-null  int64
 55  calendar_last_scraped                           96651 non-null  str
 56  number_of_reviews                               96651 non-null  int64
 57  number_of_reviews_ltm                           96651 non-null  int64
 58  number_of_reviews_l30d                          96651 non-null  int64
 59  availability_eoy                                96651 non-null  int64
 60  number_of_reviews_ly                            96651 non-null  int64
 61  estimated_occupancy_l365d                       96651 non-null  int64
 62  estimated_revenue_l365d                         62684 non-null  float64
 63  first_review                                    71487 non-null  str
 64  last_review                                     71487 non-null  str
 65  review_scores_rating                            71487 non-null  float64
 66  review_scores_accuracy                          71472 non-null  float64
 67  review_scores_cleanliness                       71478 non-null  float64
 68  review_scores_checkin                           71443 non-null  float64
 69  review_scores_communication                     71466 non-null  float64
 70  review_scores_location                          71442 non-null  float64
 71  review_scores_value                             71442 non-null  float64
 72  license                                         0 non-null      float64
 73  instant_bookable                                96651 non-null  str
 74  calculated_host_listings_count                  96651 non-null  int64
 75  calculated_host_listings_count_entire_homes     96651 non-null  int64
 76  calculated_host_listings_count_private_rooms    96651 non-null  int64
 77  calculated_host_listings_count_shared_rooms     96651 non-null  int64
 78  reviews_per_month                               71487 non-null  float64
dtypes: float64(21), int64(24), str(34)
memory usage: 227.0 MB
```

You should get that the data frame has a mix of `float64`, `int`, and `object` (text) columns and that some columns contain many nulls. You will also get an *estimate* of memory usage that may differ substantially from the more complete picture provided below, which suggests a 'true' value of 227MB.

4.0.3 Memory Usage

If you really need to get into the 'weeds' and profile your data frame because you are crashing Python and seeing messages about 'core dumped', or experiencing appallingly poor performance (allowing for the amount of data you've just loaded!), then `memory_usage` is the way to go:

```
df.memory_usage(index=True, deep=True)
```

```
Index                                          132
id                                          773208
listing_url                                4953174
scrape_id                                   773208
last_scraped                               1739718
                                               ...
calculated_host_listings_count              773208
calculated_host_listings_count_entire_homes 773208
```

```
calculated_host_listings_count_private_rooms     773208
calculated_host_listings_count_shared_rooms      773208
reviews_per_month                                773208
Length: 80, dtype: int64
```

You should see that the data frame uses 238,022,570 bytes of memory, but the *really* important thing to note here is the difference between `string` and other types of data: keeping data as raw strings (instead of converting to categories, for instance) uses up a *lot* more memory and this can have a huge impact on the performance of your code.

4.0.4 Printing the Columns

Finally, I find it *very* useful to be able to quickly print out a list of the **columns** without all of the details shown above. You just need to *print* the *columns* as a *list*:

```
print(df.columns.to_list())
```

```
['id', 'listing_url', 'scrape_id', 'last_scraped', 'source', 'name', 'description', 'ne
```

You should get a list showing every single column. If you get `Index(['id', 'listing_url',...], dtype='object')` then you have printed the column *index* object and you to need to tell the object to convert its output **to a list** (*hint*: Google).

5 Managing Your Data

When starting out it's common to think in terms of there being one input (the raw data) and one output (the results) to an analysis. In practice, you typically will have many intermediate outputs used as 'milestones' in the overall analysis:

- You might have a 'canonical' data file that has dealt with formatting issues and converted the columns to appropriate data types.
- You might have a 'clean' data file that has dealt with observations that seem to be incomplete or otherwise improperly formatted.
- You might generate subsets by region or area.
- You might produce an 'analytical' or 'final' data set appropriate to a specific analysis.

The purpose of the intermediate outputs is to accelerate each cycle in your development work: if you are re-running the *entire* analytics pipeline *every* time you make a small change at the tail-end of your processing pipeline then you will be drinking a lot of tea because most of your time will be spent watching the computer churn away at the data. It's also common to receive regular updates of data sets (e.g. InsideAirbnb data is updated rouhgly every six months, while Land Registry's Price Paid Data for the UK is updated monthly) and you'll need to know which release you're working with (i.e. have I already processed the September data???).

So we try to get ahead of all this by anticipating that files will be 'versioned'. I do not mean that you'll put files in Git, but that you'll need a way to track 'provenance' and 'stage of analysis' in where and how you save your data. But we also don't want to mindlessly delete the original, 'raw' data: I always prefer to keep the original data set handy since I almost always discover that there are fields I didn't realise I needed when I started my work.

So my approach to coding is usually:

1. Download the raw file and save it locally in a `data/raw` directory.
2. Load the first `nrows` of data so that I can quickly:
    - Check that the specification matches the data and select columns/rows accordingly.
    - Identify obviously invalid rows/columns and investigate further.
    - Check the code to fix data types and (where relevant) values works.
    - Write this new, smaller file ($m` << m$ and $n` << n$) out to a `data/clean` or `data/canonical` directory (depending on whether formatting the columns is so complex or takes so long on a large data set that dealing with data types needs to be separated out from actual cleaning).
    - Test out some initial ideas for further analysis.
3. Re-run the code (remove the `nrows` limit) using the full data set.

So you might end up with something a bit like this (or this could be overkill):

```
|
|--project/
|  |--code/
|     |--extract/
|     |--canonicalise/
|     |--clean/
|     |--analyse/
|  |--data/
|     |--raw/
|        |--2025/
|        |--2024/
|     |--clean/
|        |--2025/
|        |--2024/
|  |--docs/
|     |--refs/
|     |--code/
|     |--data/
|  |--conf
```

There is even a coding library called cookiecutter that tries to help research groups and companies standardise how they manage their code and data.

> ⚠ Difficulty: Moderate
>
> Although the code here is simple, the logic is not.

5.1 File Names

You should always be looking for ways to *avoid* hard-coding values that might change over time, especially those linked to the date of the data file.

In this case you might try to work out how to make it easy to update the code to download the latest file. For instance, if the file looks like `20260619-London-listings.csv.gz` then I might well specify the `url` as `{date}-{city}-listings.csv.gz` or `{year}{month}{day}-{city}-listings.csv.gz` and set up the variables that I need beforehand, possibly in a separate 'configuration' file that I read in at the start.

Using parameters makes it easier to write robust code that doesn't have unwanted side-effects. Here's a common one: you write code to download and process a file named `20251111-data.csv.gz`. You save the outputs to `clean-data.csv.gz`.

Question

What happens when your boss asks you to process `20251211-data.csv.gz`?

5.2 File Saving

Difficulty: Low

Now save the file somewhere local so that you don't have to keep downloading 40MB of compressed data every time you want to start the practical. We'll be using this data for the rest of term, so you might as well save yourself some time and bandwidth! We'll talk more about data processing pipelines over the course of the term, but I'd suggest putting this data set into a `data/raw` folder because then you can have directories like `data/clean` and `data/analytical` as you move through the process of cleaning and prepping your data for analysis.

```
path = Path(f'data/raw/{Path(url).name}') # What does this do?
print(f"Writing to: {path}")
```

```
Writing to: data/raw/20260619-London-listings.csv.gz
```

```
if not path.parent.exists(): # And what does *this* do?
    print(f"Creating {path.parent}")
    path.parent.mkdir(parents=True, exist_ok=True)

if not path.exists():
    df.to_csv(path, index=False)
    print("Done.")
```

5.3 File Loading

Now let's write something that will allow us to more quickly write our code and validate the results in exploratory phase. For simplicity I've called this 'testing', but you could also think of it as 'dev' mode. What we want is to be able to easily swap between testing and operational contexts using a 'switch' (typically, a Boolean value) and limit the data load in testing mode.

To achieve this you could set pandas to:

- Load only the first 10,000 rows using `nrows` *if* we are testing
- Use the columns specified in `cols`
- Allow pandas to load the entire data set before deciding on the column type by setting `low_memory` appropriately.

5.3.1 Row Subsetting

Let's tackle the *rows* problem first:

Question

```
testing = True

if testing:
    df = pd.read_csv(path,
                low_memory=??, ??)
else:
    df = pd.read_csv(path,
                low_memory=??)

print(f"Data frame is {df.shape[0]:,} x {df.shape[1]}")
```

So notice how this code deliberately works the same for either testing *or* operational execution – we just flip between the option by changing the `testing` variable from `True` to `False`!

To make this more robust and useful we could use this `testing` variable *throughout* our code if we wanted to change other behaviours based on development/deployment context. The state of the switch could then be set globally using an external configuration file (usually just called a 'conf file'). The easiest way to do this is to have a `conf.py` which contains your global parameters and then every script or notebook file reads in the configuration and sets these variables.

Something like:

```
testing = False
```

And:

```
from conf import *
```

5.3.2 Column Subsetting

Now let's tackle the column problem… In order to avoid having to load lots of data that we aren't sure we need yet, we can restrict the columns that we load. We got `cols` below by copying the output of (`df.columns.to_list()` and then removing the fields that we thought we *weren't* interested in.

```
cols = ['id', 'listing_url', 'last_scraped', 'name',
    'description', 'host_id', 'host_name', 'host_since',
    'host_location', 'host_about', 'host_is_superhost',
    'host_listings_count', 'host_total_listings_count',
    'host_verifications', 'latitude', 'longitude',
    'property_type', 'room_type', 'accommodates',
    'bathrooms', 'bathrooms_text', 'bedrooms', 'beds',
    'amenities', 'price', 'minimum_nights', 'maximum_nights',
```

```
    'availability_365', 'number_of_reviews',
    'first_review', 'last_review', 'review_scores_rating',
    'license', 'reviews_per_month']
print(f"Cols contains {len(cols)} columns.")
```

```
Cols contains 34 columns.
```

So let's extend our previous answer

Question

```
testing = True

if testing:
    df = pd.read_csv(path,
                low_memory=False, nrows=10000, ??)
else:
    df = pd.read_csv(path,
                low_memory=False, ??)

print(f"Data frame is {df.shape[0]:,} x {df.shape[1]}")
```

5.4 Releasing Memory

Two risks when working with Jupyter notebooks are:

1. You have run code in an order *other* than the order shown in the notebook; or
2. You have made edits to code but *not* re-run the changed code.

So you're still working from code that is no longer visible or where the a step (or five) has been missed/changed since you ran it! When that happens you can get *very* confusing issues because what you *see* doesn't square with what the computer has *executed*. To resolve this without having to re-run the entire notebook (though that can *also* be a good choice!) you might want to 'delete' the current object and re-load or re-run the relevant data or code.

```
del(df)
```

So we use `del(df)` to ensure that we aren't accidentally using the 'old' data frame. But another good reason to delete data you're no longer using is to free up memory.

6 Using Indexes

So let's start over from the saved data:

```
df = pd.read_csv(path, low_memory=False, usecols=cols)
```

When we use the `[[...]]` syntax we're taking a short-cut through the data by column (keeping all rows). The *full* syntax is `df[<row_selection>,<col_selection>]`. Only when we *don't* specify both does it then default to `df[<col_selection>]`.

To make the most of pandas you will need to get to grips with the logic than underpins this syntax. This is embedded in the idea of there being row and column indexes. These are *like* the columns `A`..`ZZ` and the rows `1`..n in Excel. As you'll have seen in the video, these aren't considered *data*, they are ways to *access the data*. Unlike Excel, while every data frame must *have* an index, in pandas you can 'promote' or 'demote' any column to be used *as* an index.

The **default row index** is just the row number and this will be created for you if you don't specify something else when you create the data frame. The **default column index** is created from a file's column names (works for many types of data) but you can also change these at any time.

6.1 Numeric Indexing

You can always access rows and columns by their **integer location** as if it is a 2D list and you want to access the 3rd to 5th columns and the 9th to 425th rows. So if you remember that `iloc` means you're using the index location that will help you to work out what's going on.

So can you read this:

```
df.iloc[
    4552:4557, # <- rows
    14:19      # <- columns
]
```

	latitude	longitude	property_type	room_type	accommodates
4552	51.51963	-0.06663	Private room in home	Private room	2
4553	51.48876	-0.13541	Entire rental unit	Entire home/apt	2
4554	51.52156	-0.11083	Entire rental unit	Entire home/apt	2
4555	51.59160	-0.02197	Private room in condo	Private room	1
4556	51.56785	-0.13472	Entire rental unit	Entire home/apt	8

We're accessing the 4552nd through 4556th rows, and the 14th through 18th columns.

6.2 Label Indexing

Where it can get confusing is when you see this:

```
df.loc[
    4552:4557,
    ['latitude','longitude','property_type','room_type','price']
]
```

	latitude	longitude	property_type	room_type	price
4552	51.51963	-0.06663	Private room in home	Private room	NaN
4553	51.48876	-0.13541	Entire rental unit	Entire home/apt	NaN

	latitude	longitude	property_type	room_type	price
4554	51.52156	-0.11083	Entire rental unit	Entire home/apt	NaN
4555	51.59160	-0.02197	Private room in condo	Private room	NaN
4556	51.56785	-0.13472	Entire rental unit	Entire home/apt	$230.00
4557	51.60389	-0.12628	Entire home	Entire home/apt	NaN

This code *seems* similar, but why does this use `loc` and not `iloc`, and why doesn't it actually return the same data??? In this case, the **index** (the numbers down the left-hand side in bold) is numeric, so we can treat it as either a *label* (which allows us to use `df.loc`) *or* a list-type index (which allows us to use `df.iloc`). So with `loc` the value `4557` is treated as a 'key' so it's retrieved, whereas with `.iloc` it's treated like a list index accessed with `range` and so *isn't* returned. It's a bit weird, but hopefully makes *some* sense now.

6.3 Non-numeric Indexes

Notice the change in indexing because 'listing_url' is no longer a column, it's the index now!

```
df.set_index('listing_url').iloc[0:3,13:18]
```

listing_url	latitude	longitude	property_type	room_type	accon
https://www.airbnb.com/rooms/264776	51.44306	-0.01948	Entire rental unit	Entire home/apt	10
https://www.airbnb.com/rooms/264777	51.44284	-0.01997	Entire rental unit	Entire home/apt	2
https://www.airbnb.com/rooms/264778	51.44359	-0.02275	Entire rental unit	Entire home/apt	4

Notice how this works differently if we specify a **non-numeric index**:

```
df.set_index('listing_url').loc[
    :, # <- Special syntax that means 'all rows' (or all columns)
    ['latitude','longitude','property_type','room_type','accommodates']
].head(3)
```

listing_url	latitude	longitude	property_type	room_type	accon
https://www.airbnb.com/rooms/264776	51.44306	-0.01948	Entire rental unit	Entire home/apt	10
https://www.airbnb.com/rooms/264777	51.44284	-0.01997	Entire rental unit	Entire home/apt	2
https://www.airbnb.com/rooms/264778	51.44359	-0.02275	Entire rental unit	Entire home/apt	4

> **Caution**
>
> It's vital that you understand how this code *works*. By which I mean *why* it does something at all, not exactly how to use `loc` and `iloc` (though that is also useful).
> `df.set_index(...)` changes the index from the default row number to another field in the data frame. This operation *returns* a new data frame with `listing_url` as its index. Because set index returned a data frame, we can simply add *another* method call (`iloc` or `loc`) on to the end of that line and *it* returns a new data frame in turn!

The fact that each operation returns a new data frame (or data series) is why you can even do this:

```
df.set_index('listing_url').iloc[0:3].latitude.mean()
```

```
np.float64(51.44316333333333)
```

7 Exploring Your Data

Let's start over *again* from the saved data:

```
df = pd.read_csv(path, low_memory=False, usecols=cols)
```

7.1 Selecting Rows

 Connections

You will want to refer to the Randomness lecture to understand how we can select the *same* random sample each time and to the session on Logic lecture to cover `NaN`s and `NA`s.

Difficulty: Low

I often like to start my EDA by simply printing out randomly-selected rows to get a feel for what's in the data. Does what I see square with what I read in the documentation? What does the `name` look like? What do I see in `last_scraped` and is it a sensible? What's the `id` field for?

```
df.sample(3)
```

	id	listing_url	last_scraped	na
9712	13133119	https://www.airbnb.com/rooms/13133119	2025-06-13	L
84814	1338573652236103568	https://www.airbnb.com/rooms/1338573652236103568	2025-06-12	N
19583	23163220	https://www.airbnb.com/rooms/23163220	2025-06-11	2

See if you can work out from the documentation (Google search time!) how to get the same 'random' sample every time you re-run this code block:

Question

```
df.sample(3, ??)
```

7.2 Dealing with NaNs and Nulls

> Difficulty: Hard.
>
> There is a *lot* going on here and you should be paying close attention.

If you really dig into the data you will see that a number of data types that aren't 'appropriate' for their contents: the id columns are floats; the dates aren't dates; there's a boolean that's not a boolean… It would be nice to fix these!

> Note
>
> I had intended to ask you to fix these by combining code from previous weeks with information provided in the lecture, but it turns out that the InsideAirbnb data set is *dirty*. There are a lot of `NaN` values and some of these are *deeply* problematic for some of the column types in pandas. There are also a number of challenges with other columns so, instead, I've opted to show you how I would clean this data as a *first pass* to get it into a format where it's tractable for further cleaning.

7.2.1 Identifying Problem Rows

The reason I'm not asking you to do this part yourselves is that it took me nearly an hour just to work out why I couldn't convert some of the columns to the right data types; then I started finding rows like these:

```
df[df.price.isna()][['id','name','price','room_type']].head(4)
```

	id	name	price	room_type
12	271122	Heathrow BNB - Home Away From Home!	NaN	Private room
13	425143	luxury 1bed in Chelsea Bridge Wharf	NaN	Entire home/apt
16	427229	Kitchener House!	NaN	Private room
29	38610	CHARMING FAMILY HOME	NaN	Entire home/apt

```
df[df.room_type.isna()][['id','name','price','room_type']].head(4)
```

id	name	price	room_type

```
df[~(df.price.str.startswith('$', na=False))][['id','name','price','room_type']].hea
```

	id	name	price	room_type
12	271122	Heathrow BNB - Home Away From Home!	NaN	Private room
13	425143	luxury 1bed in Chelsea Bridge Wharf	NaN	Entire home/apt
16	427229	Kitchener House!	NaN	Private room
29	38610	CHARMING FAMILY HOME	NaN	Entire home/apt

If I had to guess, I'd say that it's the result some kind of partial extract/write process because there *are* elements in some of the problem row(s) that look right but they are in the wrong columns. So we can *probably* drop some of these rows, but one thing to do is look at the frequency of NaNs across the data frame *first*. So we need to look for NaNs and Nulls, but it's quite obvious that a `NaN` in the listing id is a basic problem and we should drop these.

```
df[df.id.isna()][['id','name','price','room_type']]
```

id	name	price	room_type

As always, if you don't know that's going on, break it down:

- You have seen how column works (`[[<column names>]]`), so that's just selecting the columns that we want to show;
- You know how row selection works (`df[<selection criteria>]`), so that isn't anything really new either;
- So the only really new part is `df.id.isna()`: `df.id` is the `id` column (we could have written this `df['id']` if we wanted) and `isna()` is a test for whether or not a value is NaN.

So this shows that only one row in the 10,000 row sub-sample has a NaN for its id.

If you're not sure what the next line does, try breaking it down by running the inner bits before you run the `drop` command; and also try looking online for examples of how to use `df.drop` (e.g. just up above):

```
print(f"Data frame contains {df.shape[0]:,} rows.")
df.drop(df[df.id.isna()].index.array, axis=0, inplace=True)
print(f"Data frame contains {df.shape[0]:,} rows.")
```

```
Data frame contains 96,651 rows.
Data frame contains 96,651 rows.
```

With that really troublesome data out of the way, you can now turn to counting NaNs or Nulls in the remaining data with a view to identifying other rows that can probably be dropped.

7.2.2 Counting Nulls by Column

As a starting point I would look to drop the columns that contain only NaNs. Remember that we've dropped a row from the data frame so our maximum is now $n - 1$)! Notice how this next command works:

```
# returns a data frame with all values set to True/False according to Null status
df.isnull()
# counts these values by column (we'll see another option in a moment)
df.isnull().sum(axis=0)
# Sort results in descending order
df.isnull().sum(axis=0).sort_values(ascending=False)
```

```
df.isnull().sum(axis=0).sort_values(ascending=False)[:12]
```

```
license                 96651
host_about              47337
beds                    33975
price                   33967
bathrooms               33921
first_review            25164
reviews_per_month       25164
last_review             25164
review_scores_rating    25164
host_location           23328
bedrooms                12580
description              2845
dtype: int64
```

The most obvious ones here are: license, host_about, beds, price.

```
df.drop(columns=['license','host_about'], inplace=True)
```

Because we have dropped everything `inplace` the code simply runs and doesn't return anything.

7.2.3 Counting Nulls by Row

We now know that there *are* still quite a few problems, but we do still need a way to identify the rows that are causing most of the problems.

Notice here that the change from `axis=0` to `axis=1` changes the 'direction' of the `sum` from columns to rows. And we are getting back a data series because the summing operation reduces it to just one column.

```
df.isnull().sum(axis=1).sort_values(ascending=False).head(10)
```

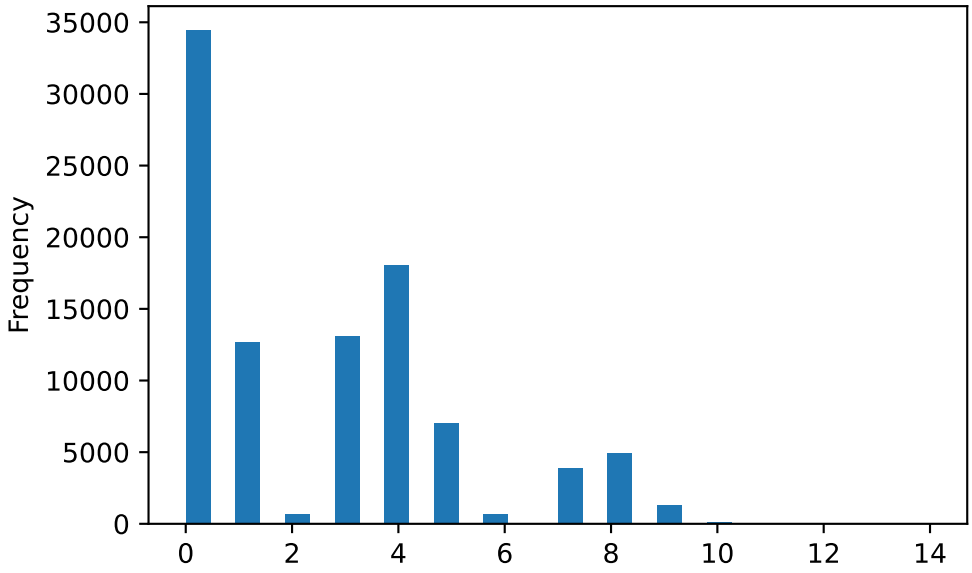
```
3646     14
5155     14
5535     14
4622     14
6122     14
30641    13
31005    11
627      11
678      10
34449    10
dtype: int64
```

So that `Series` shows how many `NaN` values there are by index value (i.e. the row number). The first column is the row id, the second is the number of NaNs in that row.

If we save the results to a variable called `probs` (i.e. problems) then we can decide what to do next.

```
probs = df.isnull().sum(axis=1)
print(type(probs))       # Note that this has returned a series!
probs.plot.hist(bins=30) # Oooooooh, check out what we can do with a series!
```

```
<class 'pandas.Series'>
```



Looking at this histogram, I would think about dropping rows missing more than about 5 values on the basis that they are the ones that are most likely be problematic. We can use the index from `probs` to select out the rows we want to inspect from the main data frame.

Here's another bit of code that bears unpacking:

```
print(f"df contains {df.shape[0]:,} rows.")
cutoff = 5
df.drop(probs[probs > cutoff].index, inplace=True)
print(f"df contains {df.shape[0]:,} rows.")
```

```
df contains 96,651 rows.
df contains 85,816 rows.
```

1. `probs > 5`: this selects only those rows in the 'probs' series whose value is greater than 5
2. `probs[...].index` returns the index values from the Series, which we will then pass to the `drop` command.
3. `df.drop(..., inplace=True)` will then drop the rows selected by `probs[probs>5].index`.

8 Fixing Data Types

When we fix the data types we are undertaking a kind of data 'profiling': working out what kind of data we are working with and how it should be represented at the level of observations and

columns. There are *huge* benefits to computer memory and diskspace usage to profiling and, consequently, huge gains to be made in the speed of data analysis.

If you want to challenge yourself, then I'd suggest trying to work out how to adapt what we saw in previous weeks using the data type dictionary to map column names to column types; however, a more straightforward way to do this is to create different for loops for each.

8.1 Profiling (Not Supported)

> Difficulty: Low.

The Pandas Profiling tool (rebranded a year or so back as ydata-profiling) offers an alternative way of understanding what's going on in your data. The output looks rather nice and you might be tempted to ask why we didn't use this straight away on the full data set – well, if you really want to know, see what happens when you profile all 70,000-odd rows and 70-odd columns in the raw data frame… in effect: while it's 'nice to have', the likelihood of crashing your computer increases significantly and it's a bit of a tangent, so that's why it's no longer included in the Podman image.

If you *do* want to explore this then you'll need to install the library, and **this is a good chance to look at how to install software on another machine**:

```
from ydata_profiling import ProfileReport
```

8.2 Managing Memory

> Difficulty: Low.

So as to *why* you'd want to fix your data types, there are two reasons: 1) to ensure that you can make the *most* of your data; 2) to ensure that it takes up as little space as possible in memory. Some simple examples:

- A column containing only the strings `'True'` (4 bytes) and `'False'` (5 bytes) will take up vastly more space than a column containing only `True` and `False` (1 **bit** each).
- A column containing only `'Red'`, `'Green'`, and `'Blue'` (3, 5, and 4 bytes each respectively) will take up much more space that a column where we use the numbers `1, 2, 3` to represent these values and have a map that tells us `1==Red`, `2==Blue`, and `3==Green`.

Let's test this idea out before looking more closely at how to convert each type of data:

```
# String type memory usage
rtm = df.room_type.memory_usage(deep=True)
# Categorical type memory usage
ctm = df.room_type.astype('category').memory_usage(deep=True)

print(f"The raw memory usage of `room_type` is {rtm/1024:,.0f} Kb.")
print(f"The categorical memory usage of `room_type` is {ctm/1024:,.0f} Kb.")
print(f"That's {(ctm/rtm)*100:.0f}% of the original!")
```

```
The raw memory usage of `room_type` is 2,524 Kb.
The categorical memory usage of `room_type` is 754 Kb.
That's 30% of the original!
```

```
# String type memory usage
shm = df.host_is_superhost.memory_usage(deep=True)
# Boolean type memory usage
bhm = df.host_is_superhost.replace({'f':False, 't':True}).astype('bool').memory_usag

print(f"The raw memory usage of `host_is_superhost` is {shm/1024:,.0f} Kb.")
print(f"The boolean memory usage of `host_is_superhost` is {bhm/1024:,.0f} Kb.")
print(f"That's {(bhm/shm)*100:.0f}% of the original!")
```

```
The raw memory usage of `host_is_superhost` is 1,434 Kb.
The boolean memory usage of `host_is_superhost` is 754 Kb.
That's 53% of the original!
```

8.3 Boolean Values

⚠ Difficulty: Moderate.

Let's start with columns that are likely to be boolean:

```
bools = ['host_is_superhost']
df.sample(5, random_state=43)[bools]
```

	host_is_superhost
73503	f
30837	f
50741	f
46926	f
37637	t

Here we have to `map` 't' to True and 'f' to False *before* converting the column to a boolean type. If you simply tried to replace them with the strings 'True' and 'False', then any string that is not `None` would convert to a `True` boolean.

```
# This approach requires us to map 't'
# and 'f' to True and False
for b in bools:
    print(f"Converting {b}")
    df[b] = df[b].replace({'f':False, 't':True}).astype('bool')
```

```
Converting host_is_superhost
```

```
df.sample(5, random_state=43)[bools]
```

	host_is_superhost
73503	False
30837	False
50741	False
46926	False
37637	True

8.4 Dates

Difficulty: Hard.

I've found dates to be particularly challenging, though pandas has *tried* to make this process less painful than it was a few years ago. What can be particularly frustrating is if *one* row has a non-sensical date value (e.g. a `t`, as happened in 2019/20) then the entire type conversion will fail. When that happens, pandas is not great about communicating where the problem occurred and I had to work it out by trying to convert *parts* of each series (using `.iloc`) to the datetime type until I had a block that failed. I then knew that I could narrow this down further using integer location indexing.

```
dates = ['last_scraped','host_since','first_review','last_review']

print(f"Currently {dates[1]} is of type '{df[dates[1]].dtype}'", "\n")
df.sample(5, random_state=43)[dates]
```

```
Currently host_since is of type 'str'
```

	last_scraped	host_since	first_review	last_review
73503	2025-06-12	2016-11-29	2024-10-20	2025-05-05
30837	2025-06-13	2011-05-11	2019-12-01	2020-03-08
50741	2025-06-12	2017-11-17	2023-05-13	2023-12-30
46926	2025-06-12	2021-07-30	2023-02-27	2025-05-25
37637	2025-06-13	2018-06-07	NaN	NaN

```
for d in dates:
    print("Converting " + d)
    df[d] = pd.to_datetime(df[d])
```

```
Converting last_scraped
Converting host_since
Converting first_review
Converting last_review
```

```
df.sample(5, random_state=43)[dates]
```

	last_scraped	host_since	first_review	last_review
73503	2025-06-12	2016-11-29	2024-10-20	2025-05-05
30837	2025-06-13	2011-05-11	2019-12-01	2020-03-08
50741	2025-06-12	2017-11-17	2023-05-13	2023-12-30
46926	2025-06-12	2021-07-30	2023-02-27	2025-05-25
37637	2025-06-13	2018-06-07	NaT	NaT

Of course, it's not actually clear there what has changed! But if you dig a little more deeply:

```
print(f"Now {dates[1]} is of type '{df[dates[1]].dtype}'", "\n")
df.sample(5, random_state=45)[dates[1]].dt.strftime('%A %B %d, %Y')
# Try some other formats!
```

```
Now host_since is of type 'datetime64[us]'


55151    Tuesday December 27, 2011
17993    Wednesday August 10, 2011
37819       Thursday June 07, 2018
2994      Monday December 01, 2014
41969    Friday September 04, 2015
Name: host_since, dtype: str
```

In that line of code we:

- Took a random sample (setting the state to 45),
- Took the second column from the dates list (`dates[1]`),
- Used the *date* 'accessor method' (`.dt`),
- And called `string format time` with the format `%A %B %d, %Y` (Full Day of Week, Month Name, Date, 4-digit Year)

8.5 Categories

> ⚠ Difficulty: Moderate.

We know that these are likely to be categories because there'd be no other way to allow users to effectively search Airbnb.

```
cats = ['property_type','room_type']

print(f"Currently {cats[1]} is of type '{df[cats[1]].dtype}'", "\n")
df.sample(5, random_state=42)[cats]
```

```
Currently room_type is of type 'str'
```

	property_type	room_type
61872	Entire home	Entire home/apt
3813	Private room in home	Private room
17706	Entire home	Entire home/apt
63429	Private room in rental unit	Private room
73215	Entire home	Entire home/apt

This next piece of code is quite useful for grouping and counting operations: we are counting the occurences of each unique value in part particular column or combination of columns:

```
df[cats[0]].value_counts()
```

```
property_type
Entire rental unit                    37143
Private room in rental unit           11610
Private room in home                  10121
Entire condo                           8113
Entire home                            7848
                                      ...
Private room in shipping container        1
Cave                                      1
Lighthouse                                1
Private room in resort                    1
Shared room in loft                       1
Name: count, Length: 88, dtype: int64
```

```
df[cats[1]].value_counts()
```

```
room_type
Entire home/apt    57168
Private room       28367
Shared room          180
Hotel room           101
Name: count, dtype: int64
```

> **Tip**
>
> One column has *many* different values (including Campers/RVs and Yurts!), the other has just four. If I were looking to conduct research I'd probably *start* with the `room_type` column since I may not care about hotels and therefore never even need to decide whether I care about boutique ones!

```
for c in cats:
    print(f"Converting {c}")
    df[c] = df[c].astype('category')
```

```
Converting property_type
Converting room_type
```

```
print(f"Now {cats[1]} is of type '{df[cats[1]].dtype}'", "\n")
print(df[cats[1]].cat.categories.values)
```

```
Now room_type is of type 'category'

<ArrowStringArray>
['Entire home/apt', 'Hotel room', 'Private room', 'Shared room']
Length: 4, dtype: str
```

```
df.sample(5, random_state=42)[cats]
```

	property_type	room_type
61872	Entire home	Entire home/apt
3813	Private room in home	Private room
17706	Entire home	Entire home/apt
63429	Private room in rental unit	Private room
73215	Entire home	Entire home/apt

8.6 Dealing with Strings

Difficulty: Hard.

We'll have to put some more work into dealing with the description and other 'free-from' text fields later in the term, but for now let's just deal with a straightforward one: price!

```
money = ['price']
df.sample(5, random_state=42)[money]
```

	price
61872	NaN
3813	$43.00
17706	NaN
63429	$65.00
73215	NaN

You will get an error when you run the next code block, that's because I want you to do a little thinking about how to extend the code to fix the data. You've already got the code you need to fix it, you just need to do a bit of thinking about 'method chaining'!

```
for m in money:
    print(f"Converting {m}")
    try:
        df[m] = df[m].str.replace('$','', regex=False).astype('float')
    except ValueError as e:
        print(f"    xxxx Unable to convert {m} to float xxxx")
        print(e)
```

```
Converting price
    xxxx Unable to convert price to float xxxx
could not convert string to float: '1,500.00'
```

Look closely at the error and then think about what you need to add to the code below:

> **Note**
>
> For now don't worry about what `regex=False` means. It will all make sense when we get to *dealing with text*.

Question

```
for m in money:
    print(f"Converting {m}")
    df[m] = df[m].str.replace('$','', regex=False).str.replace(??).astype('float')
```

```
df.sample(5, random_state=42)[money]
```

	price
61872	NaN
3813	43.0
17706	NaN
63429	65.0
73215	NaN

And here's a final thing to note that looks… a little odd:

```
df.sort_values(by='price', ascending=False).head(5)[['id','name','price','minimum_ni
```

	id	name	price	minimum_
10262	13841484	Bright & airy DoubleBed with EnSuite in Zone 2!	74100.0	2
56678	957005187369596707	Close To London Eye	65000.0	2
82485	1311151886101957046	Walk To London Eye	58000.0	2
84270	1334032580590137938	Amazing Central Location (NBOL)	58000.0	2
59371	1009450636308513568	Close to London Eye (BOL)	58000.0	2

8.7 Dealing with Integers

Difficulty: Hard.

This is the issue that made me abandon the idea of making you clean the data yourselves. Although *floats* have no issues with `np.nan` in the Series, by default there are no numpy integer arrays that can cope with NaNs. This was such a major issue for Pandas that they've actually created their *own* data type that *does* support NaN values in integer columns. There are a lot of integer columns, but only one of them seems to be a problem.

```
ints  = ['id','host_id','host_listings_count','host_total_listings_count','accommoda
         'beds','minimum_nights','maximum_nights','availability_365']
for i in ints:
    print(f"Converting {i}")
    try:
        df[i] = df[i].astype('float').astype('int')
    except ValueError as e:
        print("  - !!!Converting to unsigned 16-bit integer!!!")
        df[i] = df[i].astype('float').astype(pd.UInt16Dtype())
```

```
Converting id
Converting host_id
Converting host_listings_count
Converting host_total_listings_count
Converting accommodates
Converting beds
  - !!!Converting to unsigned 16-bit integer!!!
Converting minimum_nights
Converting maximum_nights
Converting availability_365
```

So we convert the column but using a `try / except` approach that allows to trap `ValueError` exceptions triggered by the presence of NaNs in the column. The following code tells us that there are just eight of these in the 10k sample, but they're enough to cause the code to fail if you don't trap them. The alternatives would be to: a) drop those rows; or b) leave the data as floats. For some reason the latter offends my sense of order, and the former feels like avoiding the problem rather than dealing with it.

```
df.beds.isna().value_counts()
```

```
beds
False    62065
True     23751
Name: count, dtype: int64
```

8.8 Validation

> Difficulty: Low.

Ordinarily, at this point I would then output information to confirm that all of the opeations I *think* I've undertaken were correctly applied.

```
df.info()
```

8.9 Saving

Also at this point I would save a copy of the cleaned data, though I would only consider this data *partially* cleaned since we've not made it any further than just ensuring that each column is in an appropriate format and that some particularly problematic rows have been dropped!

```
csv_out = Path(f'data/clean/{path.name}')
pq_out  = Path(f'data/clean/{path.name.replace('.csv.gz','.parquet')}')

if not csv_out.parent.exists():
    print(f"Creating {csv_out.parent}")
    csv_out.parent.mkdir(parents=True, exist_ok=True)

df.to_csv(csv_out, index=False)
df.to_parquet(pq_out, index=False)
print(f"Saved {df.shape[0]:,} rows of {df.shape[1]:,} columns to {csv_out.resolve()}")
print("Done.")
```

```
Saved 85,816 rows of 32 columns to /home/jovyan/work/practicals/data/clean/20260619-London-listings.csv.gz
Done.
```

We'll shortly begin to look at the `parquet` file format because it's fast, it preserves data types, it's compressed, and it will avoid the kinds of the problems that come up when you move to/from CSV as a default; however, for now let's keep working with what we understand.

9 Selection using Criteria

So far we've been taking primarily a row and column view of the data, now we want to think more formally about selecting ranges from within the data set…

9.1 Selecting using Data Types

> Difficulty: Low.

If we wanted to filter in/out certain columns pandas can do that! Let's try for floats and ints (*hint*: these are 64-bit data types).

Question

```
df.select_dtypes(include=[??])
```

9.2 Selecting using Conditions

> Difficulty: Hard.

Conditional selection is usally done as a combination of the selection approaches above in combination with conditionals. So to try to select only the `Entire home/apt` room type we are testing for cases where the `room_type` equals our target term (`Entire home/apt`):

Question

```
df[df.??=='??']['property_type'].value_counts().head(10)
```

Your output should be:

```
property_type
Entire rental unit           37143
Entire condo                  8113
Entire home                   7848
Entire serviced apartment     1794
Entire townhouse               989
Entire loft                    313
Entire guesthouse              218
Name: count, dtype: int64
```

9.3 Arbitrary Selection Criteria

> Difficulty: Moderate, if the previous section made sense to you.

OK, now let's look for the Entire home/apt listings that cost more than the average price of all listings… to do *that* let's get a sense of where the mean and median value fall:

Question

```
print(f"The mean price is ${df.price.??():0.2f}")
print(f"The median price is ${df.price.??():0.2f}")
```

You should get:

- The mean price is $211.15
- The median price is $134.00

You should see that the mean is higher than the median price but both are *very* roughly plausible values. Given your understanding of distributions from, say, Quantitative Methods, what can you say about the pricing distribution of Airbnb units?

You might want to have a look at the documentation: it's rather a long list, but most of your descriptive stats are on that page in the Cumulative / Descriptive Stats section, and there's also lots of information about methods for strings and categorical data.

9.3.1 Filtering: it's 'logical'

So we want to take `Entire home/apt` and filter the data set *together with* the price per night from the `price` column. For that, let's use the mean price/night of $211.15. *Note*: this is totally arbitrary.

Question

So here we want to filter on two values in the data set using `&`:

```
pricey = df[
    (??) &
    (df.price>df.price.??)
]
print(f"Selected {pricey.shape[0]:,} rows")
```

You should get 16,220 rows.

In the code above we see two things:

1. The use of the bitwise `&` (it's *not* the same as `and` and you should recall our work with the `bitarray` earlier in the term).
2. The fact that you need parentheses around the selection in order to make the the `&` work.

9.4 Selection with an Aggregate

> Difficulty: Low.

Let's find the cheapest and most expensive listings using `min` and `max` methods:

Question

Least expensive:

```
df[df.price==df.price.??()][['price','id','listing_url','room_type','description']]
```

Most expensive:

```
df[df.price==df.price.??()][['price','id','listing_url','room_type','description']]
```

You should see one or more units priced at exceedingly high levels… and here's a way to see a few more of these budget-busting options.

```
df.sort_values(by='price', ascending=False).head(3)[
    ['price','listing_url','room_type','description']
]
```

	price	listing_url	room_type	description
10262	74100.0	https://www.airbnb.com/rooms/13841484	Entire home/apt	The aparment
56678	65000.0	https://www.airbnb.com/rooms/957005187369596707	Private room	* ZONE 1<br
82485	58000.0	https://www.airbnb.com/rooms/1311151886101957046	Private room	* ZONE 1 <b

Stop: Ask yourself if the result is *plausible*.

Question

What do you make of this result?

9.5 Selection with a Range

Difficulty: Moderate

Perhaps we aren't just looking for extremes… how about all of the properties falling within the middle of the distribution? We can ask for any abitrary quantile we like, so let's go with the 25th and 75th percentile to get the middle 50% of the data. Google how to get percentiles from pandas.

Question

```
dfr = df[
            (df.price > df.price.quantile(??)) &
            (df.price < df.price.quantile(??)) ]

print(f"Lower Quartile: {df.price.quantile(??):>6.2f}")
print(f"Upper Quartile: {df.price.quantile(??):>6.2f}")
print()
print(f"Range selected contains {dfr.shape[0]:,} rows.")
print(f"Minimum price: {dfr.price.??():>6.2f}")
print(f"Maximum price: {dfr.price.??():>6.2f}")
```

That example contains a few things to which you need to pay attention:

1. *Again* you can see that, with mutiple selections, we had to put parentheses around each one – this forces Python to…
2. Process the `&` (bit-wise AND) that asks pandas to "Find all the rows where condition 1 *AND* condition 2 are both `True`". So it calculates the `True`/`False` for the left side and the `True`/`False` for the right side of the `&`, and then combines them.

I find this parentheses business annoying and frequently get an error when I forget to add them, but I'm guessing it's tied to operator precedence and how the various operations are interpreted by Python.

10 Deriving New Variables

> Difficulty:

Let's try calculating several derived measures of distribution for the price… these deliberately demonstrate different ways of handling this process (and notice also the little call to `apply` that can perform additional tasks).

10.0.1 The *z*-Score

The z-score is given by $z = (x - \bar{x})/\sigma$.

Question

```
df['z'] = (df.?? - df.??.??()) / df.??.??()
df.z.describe().apply(lambda x: f"{x:5.5f}")
```

10.0.2 Inter-Quartile Standardisation

The IQR-standardised score is given by $i = (x - Q_1)/(Q_3 - Q_1)$

Question

```
df['iqs'] = (df.price - ??)/(??-??)
df.iqs.describe().apply(lambda x: f"{x:5.5f}")
```

10.0.3 Log-Normalisation

The natural log of the price is gven by $ln(x)$

Question

```
df['lnprice'] = np.log(??)
df.lnprice.describe().apply(lambda x: f"{x:5.5f}")
```

11 Quick (and Dirty) Plotting

Although we've spent a lot of time grappling with pandas code and cleaning/filtering/selecting data, one of the first things we should really do when exploring a new dataset is plot (aka graph) the data. We've left plotting until late in this practical so that we could see some other basic attributes of how pandas stores data. We'll look at plotting and exploratory data analyses in much more detail across the following weeks, including using packages other than pandas.

For now, let's look at the basic plotting functionality pandas provides - in conjunctions with the online documentation for both DataFrames and Series. There are also examples of all the different types of plots pandas can produce.

> ⚠ MacOS plotting *without* Podman
>
> MacOS users who are *not* using Podman will need to do certain things in a specific order at the start of any notebook in order to show maps or graphs. Please make a copy of the following code for any notebook that you create and make it the *first* code that you run in the notebook…
>
> ```
> # Needed on a Mac
> import matplotlib as mpl
> mpl.use('TkAgg')
> %matplotlib inline
> import matplotlib.pyplot as plt
> ```

11.0.1 Histograms

> 💡 Difficulty: Low

First, let's see some of the ways we could visualise the distribution of the `Series` in the dataset:

```
df.price.plot.hist() # histogram
```

If the code worked properly you should have just created a standard histogram plot (if you can't see one, ask for help). However, a basic problem here may be the range of the data: if your maximum price is much more than £5,000 then you'll find the majority of your data plotted in one bar, which isn't very helpful.

You can filter the data *and* pass in some simple options to improve the plotting:

```
# Notice the ';' here to suppress `<AxesSubplot...>`
# That information doesn't *always* appear, but whenever
# you have unwanted textual output above your plot just
# add a ';' on the end of the line of code!
```

```
df[df.price < 1000].price.plot.hist(bins=50);
```

11.0.2 KDE Plots

> Difficulty: Low

Similarly, we can produce a Kernel Density Estimate plot. This time, instead of dropping data just before calling `plot` we're going to modify the *limits* of the x-axis using `xlim`:

Question

Look for information about using `xlim`:

```
df.price.plot.kde(xlim=(??)); #kernel density estimate plot
```

Kind of handy, no? These aren't the *best* looking plots, but they are all being generated on-the-fly for you by pandas with no more than a cheery `DataFrame.Series.plot.<plot type>`! Since those plots are all just method calls, many of them take optional parameters to change the colour, the notation (scientific or not), and other options. For example, many of the documentation pages linked to above are rather brief, but include a link to the general options that can be applied to all `Series.plot` calls.

This is why we like pandas: it allows us to be *constructively lazy*. We don't need to know *how* a draw a KDE plot (though it always helps if you don't see what you expected), we just need to know that pandas provides a method that will do it for you. And *that* is why it's always worth having a look at the documentation.

11.0.3 A Slight Case of Over-Plotting

Generally, Jupyter is clever enough to overlay plots one on top of the other if you call them all in the same cell. We'll see ways to gain more control later, but this is still a good start! Note that here we also need to get rid of the `-inf` values from rows that had a price of £0.

> Bug Alert
>
> The more we use pandas to sort and filter data the more you will start to see a `SettingWithCopyWarning`. This happens because of an interaction between how Pandas works and how Python works: when you are working with a very large data set you don't want to make a 'deep copy' of the data structure every time you make a change to the data. Instead, you get a 'view' into the data using a reference, which is a just a lightweight shortcut. So what happens when you try to modify that lightweight copy? Well, if you want to drop rows or columns then you either want to make a `copy()` at that point, or you will have to accept the warning *and* the computational risks that go with it.

```
# Calling copy() ensures the index is updated
# and note that all subsequent plots will have
# these £0 rows removed!
```

```
df = df[df.price > 0].copy()
df.z.plot.kde(xlim=[-2, 10])
df.iqs.plot.kde(xlim=[-2, 10])
df.lnprice.plot.kde();
```

11.0.4 Boxplots

Difficulty: Low

A standard boxplot:

```
df.lnprice.plot.box(figsize=(4, 8));
```

11.0.5 Scatterplots

Difficulty: Low

We can also plot two variables in a scatter plot by applying a plot method to the `DataFrame` (not an individual `Series`):

```
df.plot.scatter(x='longitude', y='latitude', c='price', s=2, cmap='viridis', figsize
```

Note how the code above has the form `DataFrame.plot.<plot type>`, not `DataFrame.Series.plot.<plot type>` as in the prior plots. Think about why this then means we need the `x` and `y` arguments.

Looking at the plot produced, it's hard to see where the high values are, so we might want to think about ways that we could make it easier to spot the big numbers… We could, for instance, also vary the size of the point in a plot by some variable, but why does the following not really work?

```
df.plot.scatter(x='longitude', y='latitude', c='price', s=(df.price/df.price.min()),
```

And we can plot subsets of our data without creating a new object. See if you can work out what the following code is doing that is different from the last plot:

```
df[df.price > df.price.quantile(0.90)].plot.scatter(x='longitude', y='latitude', c='
```

11.0.6 Hex Bin Plots

Difficulty: Low

And pandas allows us to create 'less standard' plots, like a hex bin plot:

```
df.plot.hexbin(x='longitude', y='latitude', gridsize=50, figsize=(10,7))
```

That's just a taste of what the basic plotting functionality of pandas can do. Feel free to explore more yourself and we'll also see the seaborn package later.

12 Credits!

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Potential Dependencies:

This notebook may depend on the following libraries: pandas, matplotlib